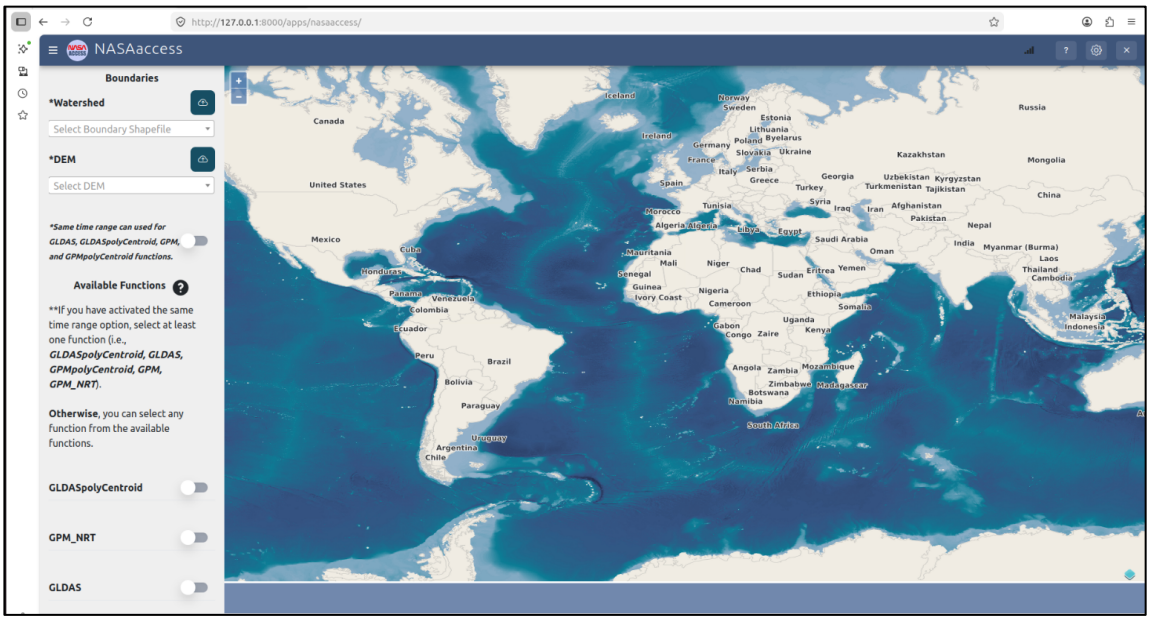
NASAaccess Tethys Web App Installation and Troubleshooting Guide for Ubuntu Linux 22.04

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Local development installation using Ubuntu 22.04 Jammy

Prepared By: Sara A. Ibrahim

Supervised by: Dr. Ibrahim N. Mohammed



This guide documents the local installation of the NASAaccess Tethys Web App on Ubuntu Linux 22.04 Jammy. It walks users through preparing the Linux system and installing all the required packages, then verifying the setup and running a small test job. The guide is intended for users who need a browser-based NASAaccess interface for uploading watershed and DEM files, connecting them to GeoServer, and accessing NASA climate and weather datasets through a local Tethys development environment. The required disk space for this is a minimum of 30GB. Using this guide you will install:

1) Python
2) Miniforge/Conda
3) Docker
4) Geoserver
5) Django
6) R
7) Tethys Platform
8) NASAaccess packages

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The installation steps are done through Ubuntu Linux, with the following specifications:
Distributor ID: Ubuntu

Description: Ubuntu 22.04.5 LTS

Release: 22.04

Codename: jammy

NASAaccess Tethys was installed on Ubuntu Linux. These commands confirm the operating system and Ubuntu release before installing packages. These lines can be pasted together, or one at a time:

```
uname -s
cat /etc/os-release
lsb_release -a
```

```
(base) internship@internship-Lenovo-V15-G4-IRU:~$ uname -s
cat /etc/os-release
lsb_release -a
Linux
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.5 LTS
Release:        22.04
Codename:       jammy
```

1 Installing Linux System Packages

Make sure everything is updated and all the required packages are installed to the latest versions.

These packages provide Git for downloading source code, curl/wget for web downloads, GnuPG and certificates for secure repositories, build tools for compiling dependencies, nano for editing files, and pigz for Docker image handling. These lines might take some time to run:

```
sudo apt update
sudo apt install -y git curl wget ca-certificates gnupg lsb-release unzip nano build-essential pigz
```

```
(base) internship@internship-Lenovo-V15-G4-IRU:~$ sudo apt update
sudo apt install -y git curl wget ca-certificates gnupg lsb-release unzip nano build-essential pigz
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:5 https://ppa.launchpadcontent.net/ettusresearch/uhd/ubuntu jammy InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
97 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.9ubuntu3).
lsb-release is already the newest version (11.1.0ubuntu4).
pigz is already the newest version (2.6-1).
ca-certificates is already the newest version (20240203~22.04.1).
curl is already the newest version (7.81.0-1ubuntu1.24).
git is already the newest version (1:2.34.1-1ubuntu1.17).
gnupg is already the newest version (2.2.27-3ubuntu2.5).
nano is already the newest version (6.2-1ubuntu0.2).
unzip is already the newest version (6.0-26ubuntu3.2).
wget is already the newest version (1.21.2-2ubuntu1.1).
0 upgraded, 0 newly installed, 0 to remove and 97 not upgraded.
```

The following lines serve as a simple check that the latest versions have been downloaded:

```
git --version
curl --version
gpg --version
```

```
(base) internship@internship-Lenovo-V15-G4-IRU:~$ git --version
curl --version
gpg --version
git version 2.34.1
curl 7.81.0 (x86_64-pc-linux-gnu) libcurl/7.81.0 OpenSSL/3.0.2 zlib/1.2.11 brotli/1.0.9 zstd/1.4.8 li
bidn2/2.3.2 libpsl/0.21.0 (+libidn2/2.3.2) libssh/0.9.6/openssl/zlib nghttp2/1.43.0 librtmp/2.3 OpenL
DAP/2.5.18
Release-Date: 2022-01-05
Protocols: dict file ftp ftps gopher gophers http https imap imaps ldap ldaps mqtt pop3 pop3s rtmp rt
sp scp sftp smb smbs smtp smtps telnet tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NT
LM NTLM_WB PSL SPNEGO SSL TLS-SRP UnixSockets zstd
gpg (GnuPG) 2.2.27
libgcrypt 1.9.4
Copyright (C) 2021 Free Software Foundation, Inc.
License GNU GPL-3.0-or-later <https://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Home: /home/internship/.gnupg
Supported algorithms:
Pubkey: RSA, ELG, DSA, ECDH, ECDSA, EDDSA
Cipher: IDEA, 3DES, CAST5, BLOWFISH, AES, AES192, AES256, TWOFISH,
        CAMELLIA128, CAMELLIA192, CAMELLIA256
Hash: SHA1, RIPEMD160, SHA256, SHA384, SHA512, SHA224
Compression: Uncompressed, ZIP, ZLIB, BZIP2
```

What-If: You get an apt lock error that looks like this?

```
Error:
Could not get lock /var/lib/dpkg/lock-frontend
It is held by process ... unattended-upgr
```

To fix, paste the following lines together:

```
ps aux | grep -E 'apt|dpkg|unattended' | grep -v grep
sudo tail -f /var/log/unattended-upgrades/unattended-upgrades.log
```

Wait until upgrades are finished. Then run:

```
sudo dpkg --configure -a
sudo apt --fix-broken install -y
sudo apt update
```

What-If: Ubuntu mirror timeout due to taking long time to install and you get 'Could not connect to ae.archive.ubuntu.com'?

Fix by switching to the main Ubuntu archive. These lines can be pasted together:

```
sudo cp /etc/apt/sources.list /etc/apt/sources.list.bak
sudo sed -i 's|http://ae.archive.ubuntu.com/ubuntu|http://archive.ubuntu.com/ubuntu|g' /etc/apt/sources.list
sudo sed -i 's|https://ae.archive.ubuntu.com/ubuntu|http://archive.ubuntu.com/ubuntu|g' /etc/apt/sources.list
sudo apt clean
sudo apt update
```

Then run the following lines:

```
sudo apt remove -y docker.io docker-doc docker-compose docker-compose-v2 podman-docker containerd runc ||
true
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
```

What-If: The computer could not translate download.docker.com into an IP address?

This is a DNS or internet connection problem. The chmod error happens only because the docker.asc file was never downloaded.

```
Error:
curl: (6) Could not resolve host: download.docker.com
chmod: cannot access '/etc/apt/keyrings/docker.asc': No such file or directory
```

Do not continue to the next Docker command until this is fixed. First, check internet and DNS by running:

```
getent hosts download.docker.com
```

If DNS is working, you should see output with IP addresses. Then run:

```
curl -I https://download.docker.com/linux/ubuntu/gpg
```

Expected result includes:

```
HTTP/2 200
```

or:

```
HTTP/1.1 200 OK
```

What-If: 'getent hosts download.docker.com' gives no output?

Restart network/DNS services, by running:

```
sudo systemctl restart systemd-resolved
sudo systemctl restart NetworkManager
```

Then test again:

```
getent hosts download.docker.com
curl -I https://download.docker.com/linux/ubuntu/gpg
```

What-If: it still fails?

Open your browser and try this page:

If Firefox also cannot open it, check the following things:

```
https://download.docker.com/linux/ubuntu/gpg
```

1) Wi-Fi/Ethernet connection
2) VPN connection
3) Campus/work network restrictions
4) Proxy settings
5) Captive portal login page

If the laptop is on an institutional network, ask IT to allow these domains:

1. download.docker.com
2. archive.ubuntu.com
3. security.ubuntu.com
4. github.com
5. conda.anaconda.org
6. repo.anaconda.com

When the test works, rerun:

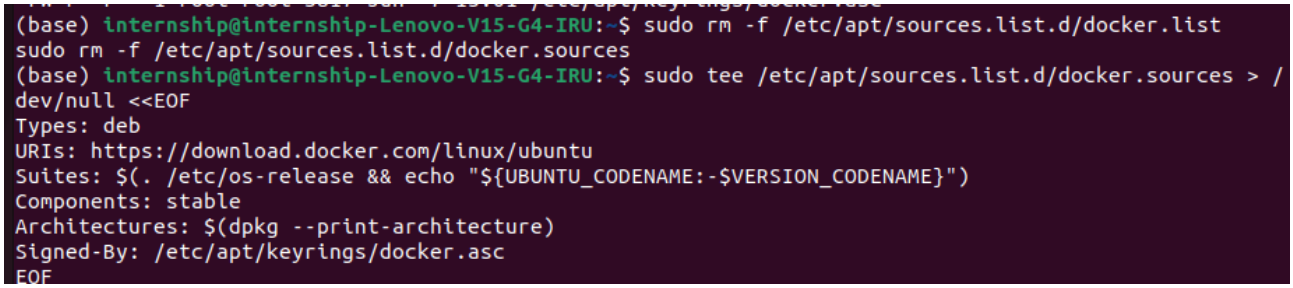
```
sudo rm -f /etc/apt/keyrings/docker.asc
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
ls -l /etc/apt/keyrings/docker.asc
```

2 Install Docker Engine

2.1 Docker Repository

Create the Docker repository source file, by pasting these lines together:

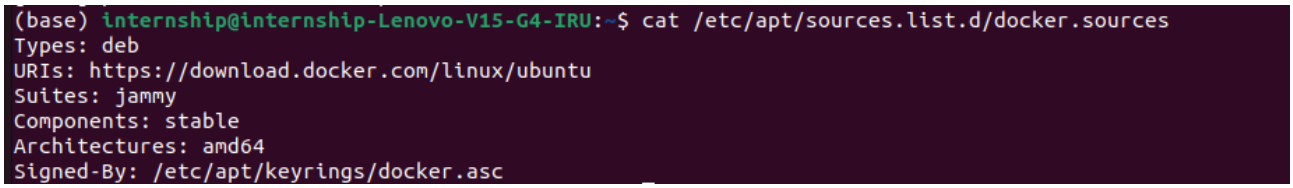
```
sudo tee /etc/apt/sources.list.d/docker.sources > /dev/null <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Architectures: $(dpkg --print-architecture)
Signed-By: /etc/apt/keyrings/docker.asc
EOF
```



Check the file, by running:

```
cat /etc/apt/sources.list.d/docker.sources
```

The expected results:



If you run the sudo apt update, below:

```
sudo apt update
```

The expected result should include:

```
Hit: https://download.docker.com/linux/ubuntu jammy InRelease
```

What-If: sudo apt update fails?

Case A – 'Could not resolve host'

Example, 'Could not resolve host: download.docker.com'

To fix it, run the following lines:

```
getent hosts download.docker.com
sudo systemctl restart systemd-resolved
sudo systemctl restart NetworkManager
getent hosts download.docker.com
sudo apt update
```

Case B – Ubuntu mirror timeout

Example, 'Could not connect to ae.archive.ubuntu.com'

Fix by switching to the main Ubuntu archive:

```
sudo cp /etc/apt/sources.list /etc/apt/sources.list.bak
sudo sed -i 's|http://ae.archive.ubuntu.com/ubuntu|http://archive.ubuntu.com/ubuntu|g' /etc/apt/sources.list
sudo sed -i 's|https://ae.archive.ubuntu.com/ubuntu|http://archive.ubuntu.com/ubuntu|g' /etc/apt/sources.list
sudo apt clean
sudo apt update
```

Case C – GPG key error

Example, 'NO_PUBKEY'

Fix:

```
sudo rm -f /etc/apt/keyrings/docker.asc
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
sudo apt update
```

2.2 Docker Installation

To install Docker, run the following line:

```
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

```
(base) internship@internship-Lenovo-V15-G4-IRU:~$ sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-ce-rootless-extras
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras
  docker-compose-plugin
0 upgraded, 6 newly installed, 0 to remove and 97 not upgraded.
Need to get 98.8 MB of archives.
After this operation, 378 MB of additional disk space will be used.
Get:1 https://download.docker.com/linux/ubuntu jammy/stable amd64 containerd.io amd64 2.2.4-1~ubuntu.22.04~jammy [23.6 MB]
Get:2 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-ce-cli amd64 5:29.5.3-1~ubuntu.22.04~jammy [16.8 MB]
Get:3 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-ce amd64 5:29.5.3-1~ubuntu.22.04~jammy [23.0 MB]
Get:4 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-buildx-plugin amd64 0.34.1-1~ubuntu.22.04~jammy [17.2 MB]
Get:5 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-ce-rootless-extras amd64 5:29.5.3-1~ubuntu.22.04~jammy [10.2 MB]
Get:6 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-compose-plugin amd64 5.1.4-1~ubuntu.22.04~jammy [8,126 kB]
Fetched 98.8 MB in 9s (10.7 MB/s)
Selecting previously unselected package containerd.io.
(Reading database ... 253369 files and directories currently installed.)
Preparing to unpack .../0-containerd.io_2.2.4-1~ubuntu.22.04~jammy_amd64.deb ...
Unpacking containerd.io (2.2.4-1~ubuntu.22.04~jammy) ...
Selecting previously unselected package docker-ce-cli.
Preparing to unpack .../1-docker-ce-cli_5%3a29.5.3-1~ubuntu.22.04~jammy_amd64.deb ...
Unpacking docker-ce-cli (5:29.5.3-1~ubuntu.22.04~jammy) ...
Selecting previously unselected package docker-ce.
Preparing to unpack .../2-docker-ce_5%3a29.5.3-1~ubuntu.22.04~jammy_amd64.deb ...
Unpacking docker-ce (5:29.5.3-1~ubuntu.22.04~jammy) ...
Selecting previously unselected package docker-buildx-plugin.
Preparing to unpack .../3-docker-buildx-plugin_0.34.1-1~ubuntu.22.04~jammy_amd64.deb ...
Unpacking docker-buildx-plugin (0.34.1-1~ubuntu.22.04~jammy) ...
Selecting previously unselected package docker-ce-rootless-extras.
Preparing to unpack .../4-docker-ce-rootless-extras_5%3a29.5.3-1~ubuntu.22.04~jammy_amd64.deb ...
Unpacking docker-ce-rootless-extras (5:29.5.3-1~ubuntu.22.04~jammy) ...
Selecting previously unselected package docker-compose-plugin.
Preparing to unpack .../5-docker-compose-plugin_5.1.4-1~ubuntu.22.04~jammy_amd64.deb ...
Unpacking docker-compose-plugin (5.1.4-1~ubuntu.22.04~jammy) ...
Setting up docker-buildx-plugin (0.34.1-1~ubuntu.22.04~jammy) ...
Setting up containerd.io (2.2.4-1~ubuntu.22.04~jammy) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /lib/systemd/system/
containerd.service.
Setting up docker-compose-plugin (5.1.4-1~ubuntu.22.04~jammy) ...
Setting up docker-ce-cli (5:29.5.3-1~ubuntu.22.04~jammy) ...
Setting up docker-ce-rootless-extras (5:29.5.3-1~ubuntu.22.04~jammy) ...
Setting up docker-ce (5:29.5.3-1~ubuntu.22.04~jammy) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /lib/systemd/system/dock
er.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /lib/systemd/system/docker.s
ocket.
Processing triggers for man-db (2.10.2-1) ...
```

Next, check Docker version:

```
docker --version
docker compose version
```

```
(base) internship@internship-Lenovo-V15-G4-IRU:~$ docker --version
docker compose version
Docker version 29.5.3, build d1c06ef
Docker Compose version v5.1.4
```

Check Docker service:

```
sudo systemctl status docker --no-pager
```

The expected result is ‘Active: active (running)’.

```
(base) internship@internship-Lenovo-V15-G4-IRU:~$ sudo systemctl status docker --no-pager
● docker.service - Docker Application Container Engine
     Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2026-06-07 16:19:38 +04; 6min ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 32835 (dockerd)
      Tasks: 18
     Memory: 25.4M
        CPU: 589ms
     CGroup: /system.slice/docker.service
             └─32835 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Jun 07 16:19:38 internship-Lenovo-V15-G4-IRU dockerd[32835]: time="2026-06-07T16:19:38.219816612…rt."
Jun 07 16:19:38 internship-Lenovo-V15-G4-IRU dockerd[32835]: time="2026-06-07T16:19:38.247918961…ges"
Jun 07 16:19:38 internship-Lenovo-V15-G4-IRU dockerd[32835]: time="2026-06-07T16:19:38.267605458…ges"
Jun 07 16:19:38 internship-Lenovo-V15-G4-IRU dockerd[32835]: time="2026-06-07T16:19:38.979328670…ne."
Jun 07 16:19:38 internship-Lenovo-V15-G4-IRU dockerd[32835]: time="2026-06-07T16:19:38.987713213….5.3
Jun 07 16:19:38 internship-Lenovo-V15-G4-IRU dockerd[32835]: time="2026-06-07T16:19:38.987765950…kit"
Jun 07 16:19:38 internship-Lenovo-V15-G4-IRU dockerd[32835]: time="2026-06-07T16:19:38.992183266…ion"
Jun 07 16:19:38 internship-Lenovo-V15-G4-IRU dockerd[32835]: time="2026-06-07T16:19:38.998870236…ion"
Jun 07 16:19:38 internship-Lenovo-V15-G4-IRU dockerd[32835]: time="2026-06-07T16:19:38.998939257…ock"
Jun 07 16:19:38 internship-Lenovo-V15-G4-IRU systemd[1]: Started Docker Application Container Engine.
Hint: Some lines were ellipsized, use -l to show in full.
```

If it is not running, run the following line:

```
sudo systemctl start docker
```

Test Docker:

```
sudo docker run hello-world
```

The expected result should include ‘Hello from Docker!’

```
(base) internship@internship-Lenovo-V15-G4-IRU:~$ sudo docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

2.3 Allow Docker without Sudo

By default, Docker commands usually require ‘sudo’ because the Docker daemon socket is owned by root. Docker’s post-installation guide says users can be added to the docker group so they can

run Docker commands without ‘sudo’, but it also warns that the docker group grants root-level privileges. Start by running:

```
sudo groupadd docker 2>/dev/null || true
sudo usermod -aG docker $USER
newgrp docker
```

Then test without ‘sudo’:

```
docker run hello-world
```

If it works, Docker is ready.

```
(base) internship@internship-Lenovo-V15-G4-IRU:~$ sudo groupadd docker 2>/dev/null || true
sudo usermod -aG docker $USER
newgrp docker
(base) internship@internship-Lenovo-V15-G4-IRU:~$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

If it says permission denied, reboot:

```
sudo reboot
```

After reboot is complete, open the CLI terminal again and run:

```
docker run hello-world
```

3 Install Miniforge / Conda

If Conda already works on your device, you skip this step and go to Step 5. Otherwise start by checking, if it is installed in your device by running:

```
conda --version
```

If it prints a version, continue to Step 5.

If Conda is missing, install Miniforge, running the following lines together:

```
cd ~/Downloads
wget "https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-$(uname)-$(uname -m).sh"
bash "Miniforge3-$(uname)-$(uname -m).sh"
```

During installation, enter yes whenever prompted to resume installation:

Accept license: yes

Install location: default

Initialize conda: yes

Close Terminal and open a new Terminal, then check:

```
conda --version
conda info --envs
```



What-If: conda --version says 'command not found'?

Try initializing Conda annually by running:

```
source ~/miniforge3/etc/profile.d/conda.sh
conda init bash
```

Then, close the terminal, open a new one and run:

```
conda --version
conda info --envs
```

You should see the downloaded versions printed as the output for these lines.

4 Create the Tethys/NASAaccess Environment

Use a separate Conda environment named Tethys. We cannot install NASAaccess into the Conda 'base' environment. We need to create a separate environment and name it 'tethys' for simplicity. Run the following lines together:

```
conda create -n tethys -c conda-forge --override-channels -y \
  python=3.11 \
  django=5.2 \
  tethys-platform=4.5.2 \
  r-nasaaccess \
  r-remotes \
  r-emayili \
  r-codetools \
  geoserver-rest \
  pandas \
  gdal \
  "pip=23.0.1" \
  "setuptools<68" \
  wheel
```

To activate the environment we just created, simply run:

```
conda activate tethys
```

To verify that the installation is complete, run:

```
python --version
python -c "import django; print(django.get_version())"
tethys version
Rscript -e "library(NASAaccess); print(packageVersion('NASAaccess'))"
which Rscript
```

The expected outcome is the Python, Django, Tethys, and NASAaccess versions being printed, respectively, followed by the Rscript path:

```
(base) internship@internship-Lenovo-V15-G4-IRU:~/Downloads$ conda activate tethys
(tethys) internship@internship-Lenovo-V15-G4-IRU:~/Downloads$ python --version
python -c "import django; print(django.get_version())"
tethys version
Rscript -e "library(NASAaccess); print(packageVersion('NASAaccess'))"
which Rscript
Python 3.11.15
5.2.15
4.5.2
[1] '4.2.0'
/home/internship/miniforge3/envs/tethys/bin/Rscript
```

What-If: Conda solving is slow or fails?

Try installing it in smaller parts:

```
conda env remove -n tethys -y
conda create -n tethys -c conda-forge --override-channels -y python=3.11
conda activate tethys
conda install -c conda-forge --override-channels -y django=5.2 tethys-platform=4.5.2
conda install -c conda-forge --override-channels -y r-nasaaccess r-remotes r-emayili r-codetools geoserver-rest
pandas gdal "pip=23.0.1" "setuptools<68" wheel
```

Then, verify again:

```
python --version
python -c "import django; print(django.get_version())"
tethys version
Rscript -e "library(NASAaccess); print(packageVersion('NASAaccess'))"
```

5 Initialize Tethys Platform

To initialize Tethys Platform, run:

```
conda activate tethys
tethys gen portal_config
```

Then configure the local database:

```
tethys db configure
```

When prompted to create an admin user, use:

Username: admin

Email: admin@example.com

Password: pass

Make sure to change the email to the email you want to link the Tethys Platform to. For a real shared machine, use a stronger password. For local use, 'admin/pass' is simple.

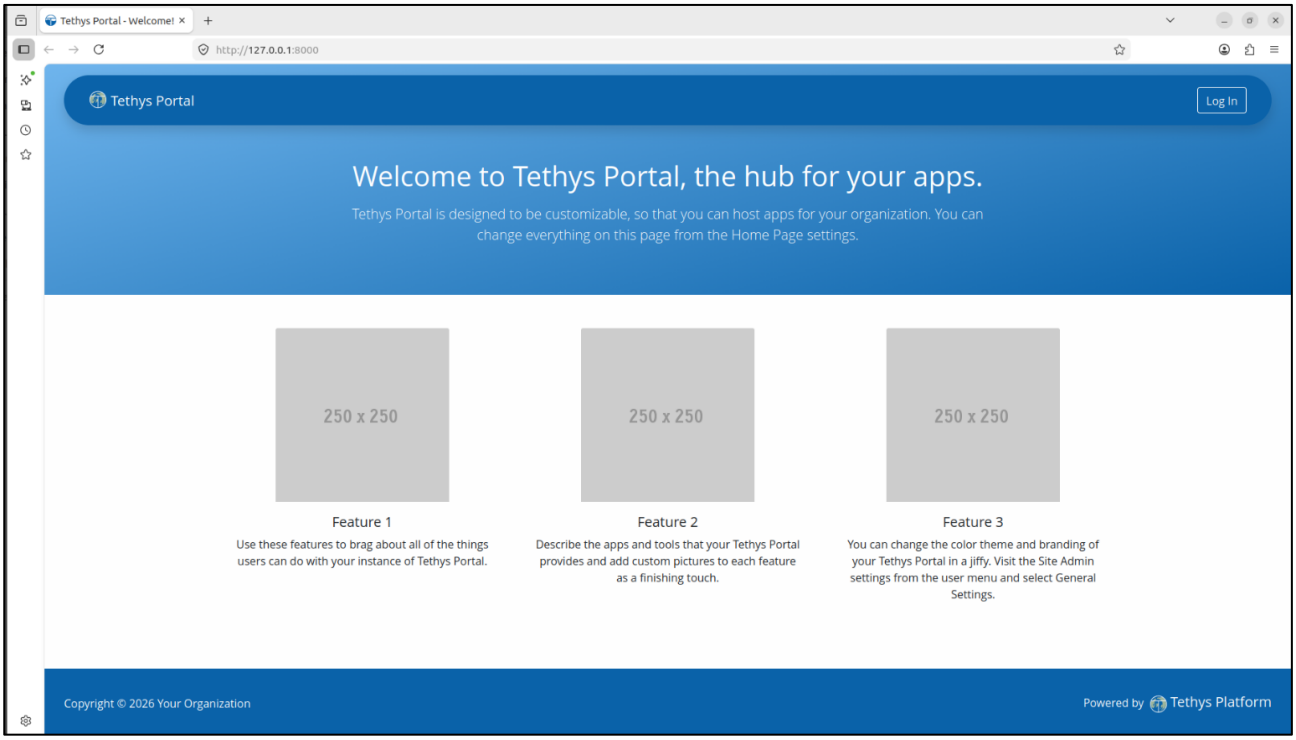
Next, start Tethys:

```
tethys start
```

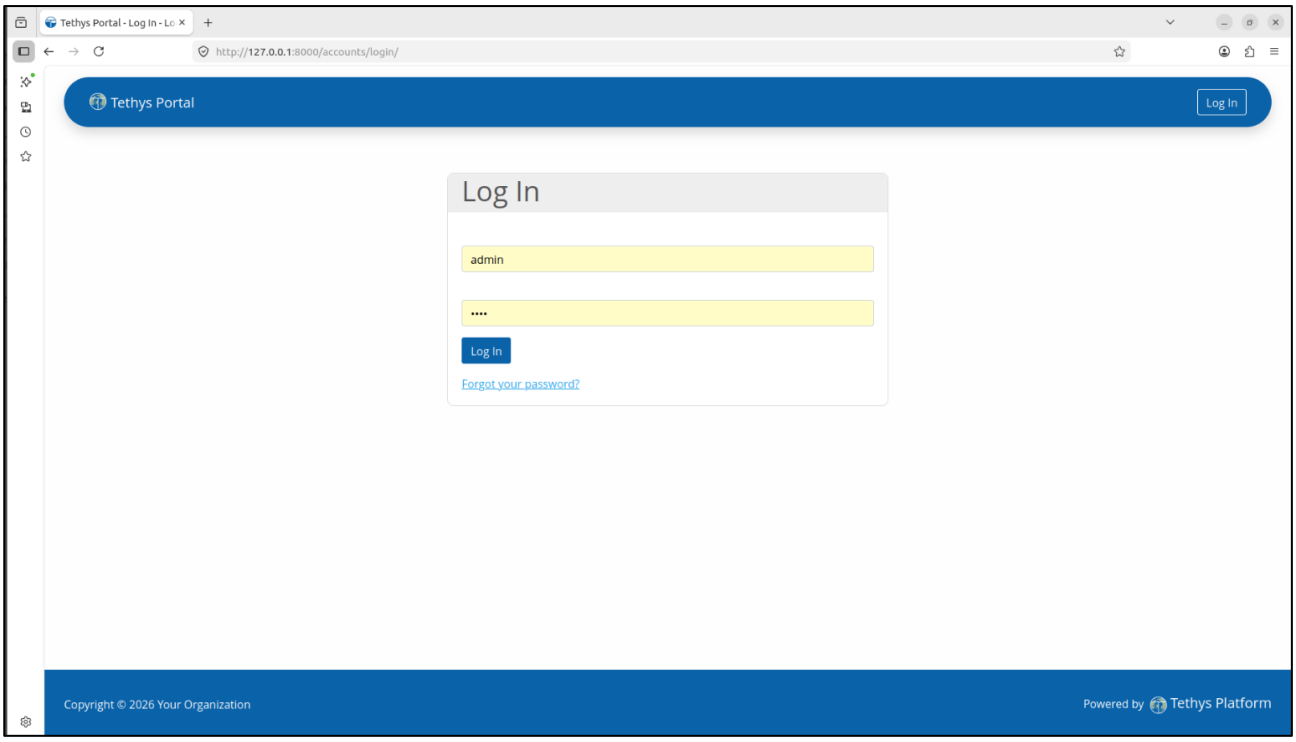
What you will see is 'http://127.0.0.1:8000/', CTRL+click on the link to open the standard Tethys Platform.



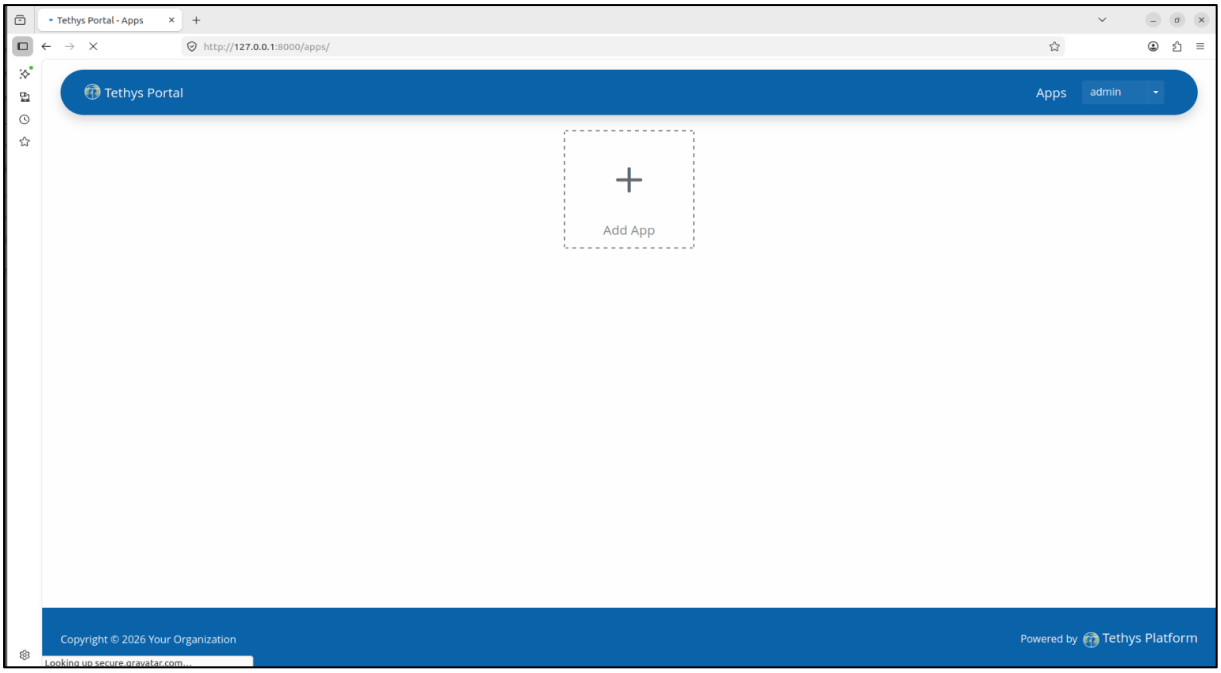
The ‘RuntimeWarning’ messages and the AXES warning shown during tethys start are not installation failures. Tethys can still start successfully if the final output says that the development server is running at http://127.0.0.1:8000/.



Login with the username: admin, and password: pass. The sign in information can be changed later from the settings.



We have not installed any web app, so the Tethys Portal will be empty.



Stop Tethys before the next step, by clicking CTRL+c

6 Install & Start GeoServer through Tethys/Docker

NASAaccess requires GeoServer for the Tethys web app, and the NASAaccess documentation shows using Tethys Docker commands to initialize and start a GeoServer container. Start by running the following two lines. The line 'conda activate tethys' is repeated in every step so that if the terminal was closed and reopened, users can continue where they left off by activating the 'tethys' environment. If the terminal was not closed and the 'tethys' environment is still activated, then this line can be ignored in the following steps.

```
conda activate tethys
tethys docker init -c geoserver
```

If it asks questions, press Enter to accept defaults.

```
(tethys) internship@internship-Lenovo-V15-G4-IRU:~/apps/tethys_nasaaccess$ conda activate tethys
tethys docker init -c geoserver
Pulling Docker images...
Digest: sha256:400a7898b81f78b7df06f6c6ed91031e0add932f8ca74d396903d19cd7f7eb56
Status: Image is up to date for tethysplatform/geoserver:latest

Installing the GeoServer Docker container...
The GeoServer docker can be configured to run in a clustered mode (multiple instances of GeoServer ru
nning in the docker container) for better performance.

Number of GeoServer Instances Enabled (max 4) [1]:
Number of GeoServer Instances with REST API Enabled (max 1) [1]:

GeoServer can be configured with limits to certain types of requests to prevent it from becoming over
whelmed. This can be done automatically based on a number of processors or each limit can be set expl
icitly.

Would you like to specify number of Processors (c) OR set request limits explicitly (e) [C/e]:
Number of Processors (max 4) [1]:
Maximum request timeout in seconds [60]:
Maximum memory to allocate to each GeoServer instance in MB (max 4096) [1024]:
Minimum memory to allocate to each GeoServer instance in MB (max 1024) [1024]:
Bind the GeoServer data directory to the host? [Y/n]:
Specify location to bind data directory [/home/internship/.tethys/geoserver/data]:

Finished installing Docker containers.
```

Start GeoServer, by running this line:

```
tethys docker start -c geoserver
```

Check by running:

```
docker ps
```

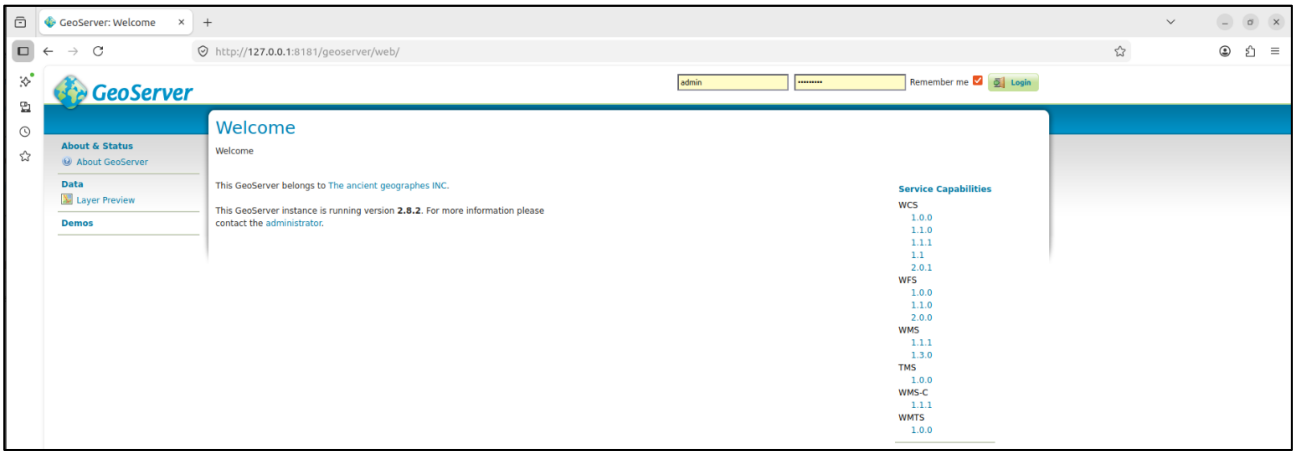
```
(tethys) internship@internship-Lenovo-V15-G4-IRU:~/apps/tethys_nasaaccess$ tethys docker start -c geo
server
Starting GeoServer container...
(tethys) internship@internship-Lenovo-V15-G4-IRU:~/apps/tethys_nasaaccess$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED              STATUS
         PORTS
                   NAMES
bb45d6ad0890   tethysplatform/geoserver:latest   "/bin/sh -c $GEOSERV…"   About a minute ago   Up 6 s
econds   0.0.0.0:8081-8084->8081-8084/tcp, [::]:8081-8084->8081-8084/tcp, 0.0.0.0:8181->8181/tcp, [::
]:8181->8181/tcp   tethys_geoserver
```

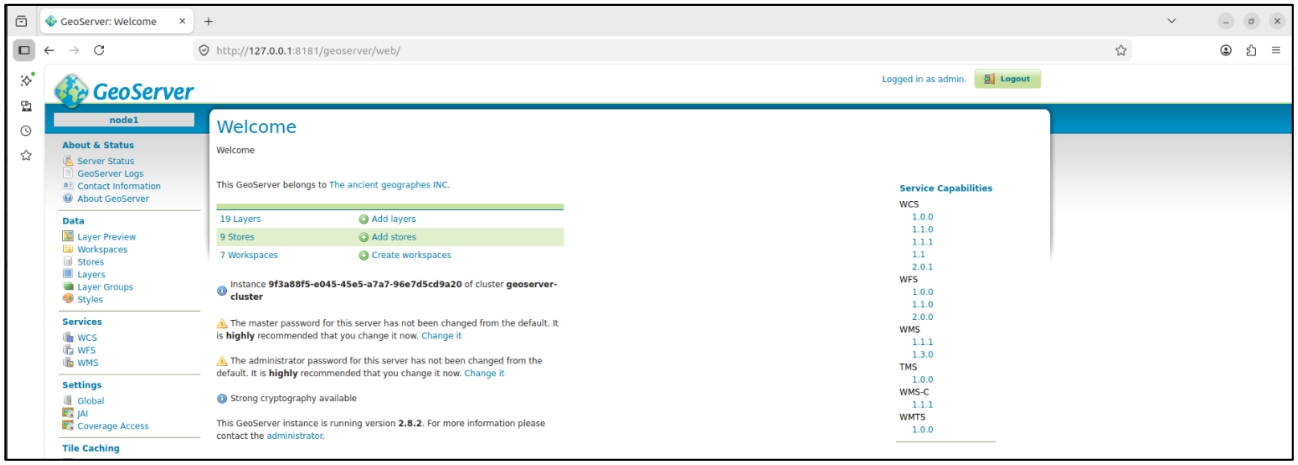
Copy and paste the following link in your browser to open GeoServer:

```
http://127.0.0.1:8181/geoserver/web/
```

Your default login information is the default, which you can be changed later:

```
Username: admin
Password: geoserver
```





What-If: GeoServer does not open?

Run:

```
docker ps
```

If no GeoServer container appears, then run:

```
tethys docker start -c geoserver
docker ps
```

Then retry the webpage link in your browser:

```
http://127.0.0.1:8181/geoserver/web/
```

If Docker permission is denied, run the following lines:

```
sudo usermod -aG docker $USER
newgrp docker
docker ps
```

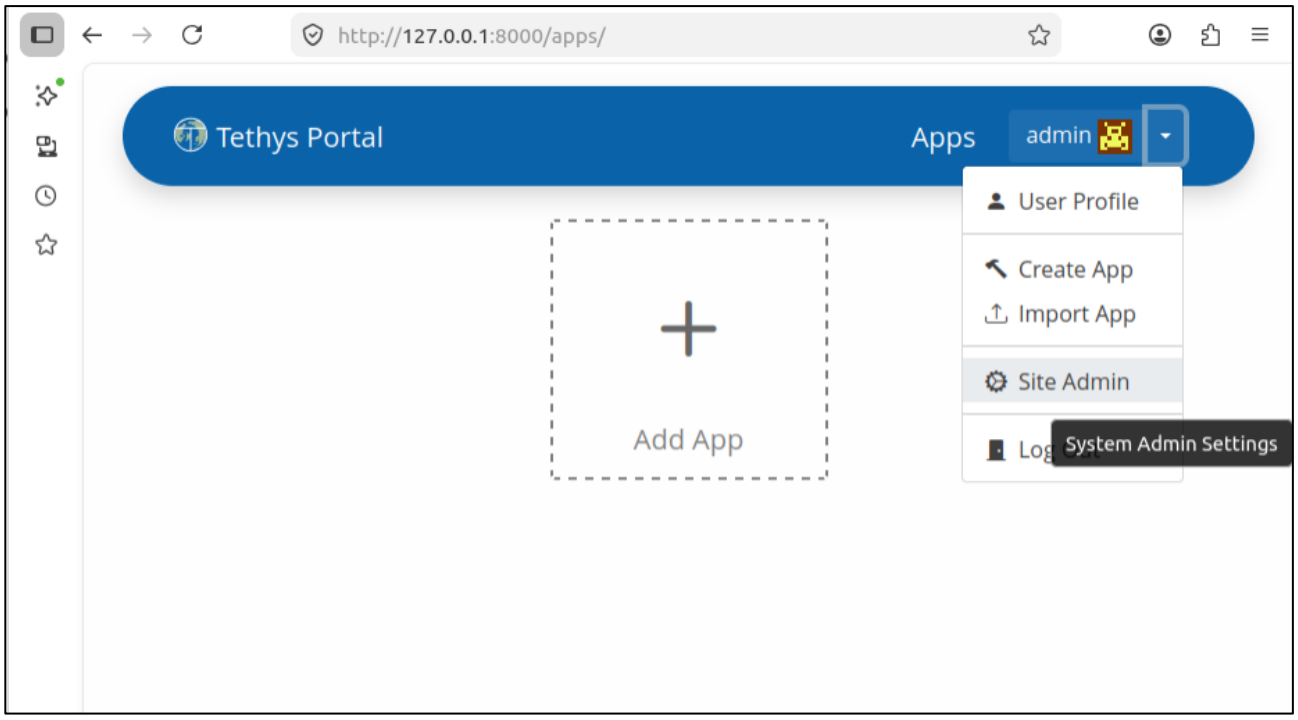
7 Register GeoServer Inside Tethys

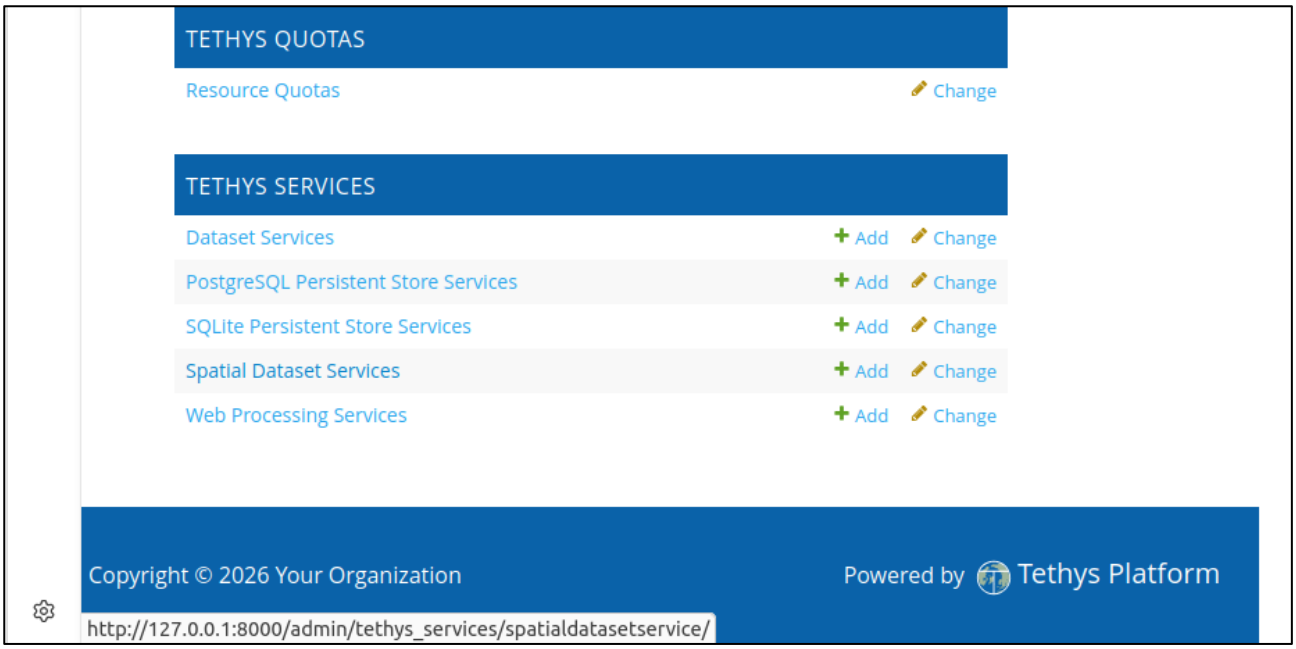
Start Tethys again:

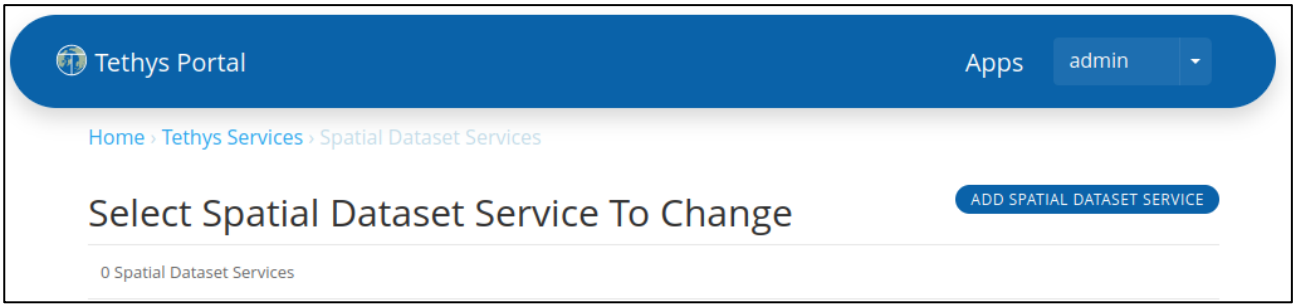
```
conda activate tethys
tethys start
```

Log in with the admin account.

Go to Site Admin → Under the last heading called 'Tethys Services' → Click on 'Spatial Dataset Services' → Click on 'Add Spatial Dataset Service' on your right.



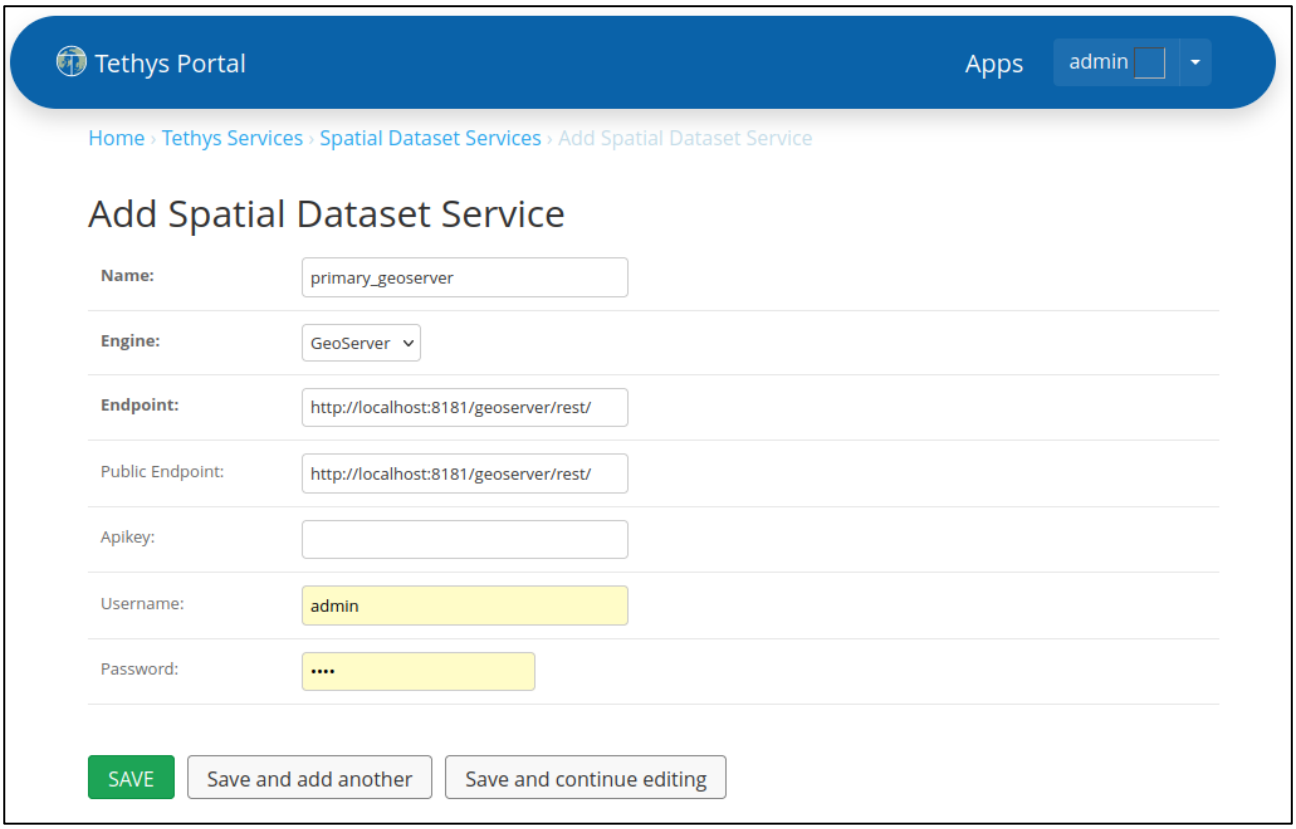




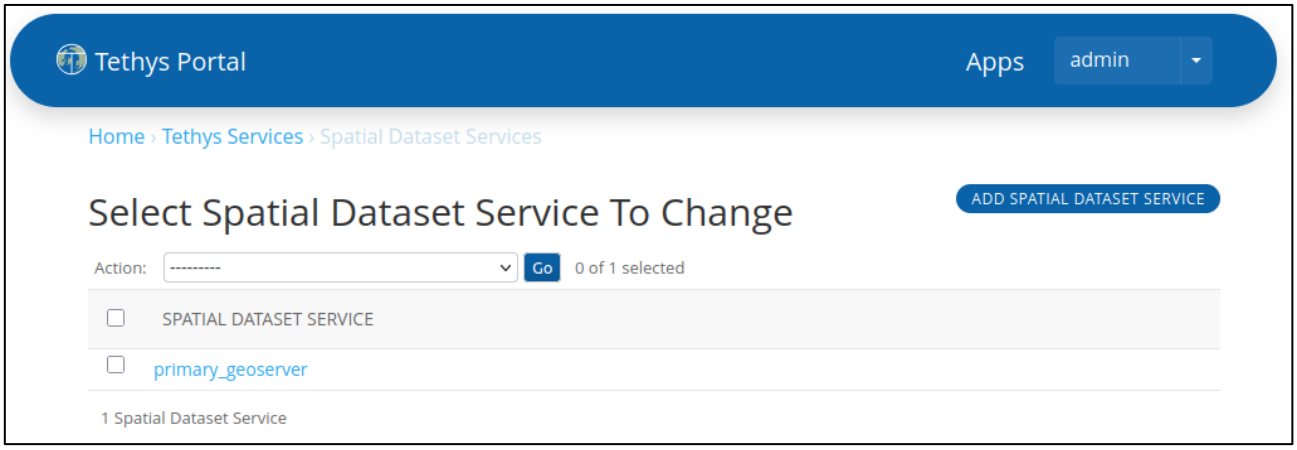
In the new spatial dataset service, type the following:

Name: primary_geoserver
Engine: GeoServer
Endpoint: http://127.0.0.1:8181/geoserver/rest/
Public Endpoint: http://127.0.0.1:8181/geoserver/rest/
Username: admin
Password: geoserver
API Key: leave blank

When entering the GeoServer password, clear the password box first as the browser autofill may insert the Tethys admin password instead of the GeoServer password. The GeoServer password must be ‘geoserver’ unless it was changed manually in the GeoServer website.



Click on save. You will see your new spatial dataset service added.



Before installing NASAaccess, verify that GeoServer REST authentication works. This catches wrong GeoServer passwords before shapefile/DEM upload. Run:

```
curl -i -u admin:geoserver http://127.0.0.1:8181/geoserver/rest/workspaces.xml | head -n 20
```

Save and stop Tethys by clicking CTRL+c in the terminal.

8 Configure NASA Earthdata Credentials

NASAaccess requires an Earthdata account, GES DISC authorization, and a '.netrc' file in the user's home directory for NASA server access. In your browser, login or register in Earthdata and record the username and password aside or memorize them.

Then create the '.netrc' file through pasting the following line:

```
nano ~/.netrc
```

The file that opens is empty. The following should be pasted inside the file, and only change the 'USERNAME' and 'PASSWORD' to your own:

```
machine urs.earthdata.nasa.gov
login USERNAME
password PASSWORD
```

Next, save the file by clicking CTRL+o, then ENTER, then CTRL+x.

Secure the files:

```
chmod 600 ~/.netrc
touch ~/.urs_cookies
chmod 600 ~/.urs_cookies
```

Check without showing the password by running:

```
ls -la ~/.netrc ~/.urs_cookies
grep "urs.earthdata.nasa.gov" ~/.netrc
```

```
(tethys) internship@internship-Lenovo-V15-G4-IRU:~$ ls -la ~/.netrc ~/.urs_cookies
grep "urs.earthdata.nasa.gov" ~/.netrc
-rw------- 1 internship docker       67 Jun 10 17:25 /home/internship/.netrc
-rw------- 1 internship internship 1422 Jun 10 17:26 /home/internship/.urs_cookies
machine urs.earthdata.nasa.gov
(tethys) internship@internship-Lenovo-V15-G4-IRU:~$
```

9 Clone the NASAaccess Tethys app

To clone NASAaccess, run the following lines:

```
conda activate tethys
mkdir -p ~/apps
cd ~/apps
git clone https://github.com/imohamme/tethys_nasaaccess.git
cd tethys_nasaaccess
mkdir -p nasaaccess_data/outputs
```

Check required files:

```
ls setup.py install.yml requirements.txt
ls tethysapp/nasaaccess/scripts/nasaaccess.R
```

What is expected to see:

```
setup.py
install.yml
requirements.txt
tethysapp/nasaaccess/scripts/nasaaccess.R
```

```
(tethys) internship@internship-Lenovo-V15-G4-IRU:~/Downloads$ conda activate tethys
mkdir -p ~/apps
cd ~/apps
git clone https://github.com/imohamme/tethys_nasaaccess.git
cd tethys_nasaaccess
mkdir -p nasaaccess_data/outputs
Cloning into 'tethys_nasaaccess'...
remote: Enumerating objects: 1242, done.
remote: Counting objects: 100% (463/463), done.
remote: Compressing objects: 100% (241/241), done.
remote: Total 1242 (delta 194), reused 347 (delta 158), pack-reused 779 (from 1)
Receiving objects: 100% (1242/1242), 16.29 MiB | 6.18 MiB/s, done.
Resolving deltas: 100% (635/635), done.
(tethys) internship@internship-Lenovo-V15-G4-IRU:~/apps/tethys_nasaaccess$ ls setup.py install.yml re
quirements.txt
ls tethysapp/nasaaccess/scripts/nasaaccess.R
install.yml  requirements.txt  setup.py
tethysapp/nasaaccess/scripts/nasaaccess.R
```

Notice that the directory changed to '~/apps/tethys_nasaaccess'. At this point you cloned the web app and its requirements from GitHub to your personal device. Next step is to install it into your Tethys.

10 Install NASAaccess into Tethys

The first step is know paths to certain folders that NASAaccess will need to reach to run analyses. Therefore, paste these lines into your Command Line Interface (CLI) and save them aside in a notepad as you will need them during the installation:

```
echo "data_path: $HOME/apps/tethys_nasaaccess/nasaaccess_data/"
echo "nasaaccess_R: $(which Rscript)"
echo "nasaaccess_script: $HOME/apps/tethys_nasaaccess/tethysapp/nasaaccess/scripts/nasaaccess.R"
echo "geoserver_workspace: nasaaccess"
echo "geoserver_URI: http://localhost/nasaaccess"
echo "geoserver_user: admin"
echo "geoserver_password: geoserver"
```

```
(tethys) internship@internship-Lenovo-V15-G4-IRU:~/apps$ echo "data_path: $HOME/apps/tethys_nasaacces
s/nasaaccess_data/"
echo "nasaaccess_R: $(which Rscript)"
echo "nasaaccess_script: $HOME/apps/tethys_nasaaccess/tethysapp/nasaaccess/scripts/nasaaccess.R"
echo "geoserver_workspace: nasaaccess"
echo "geoserver_URI: http://localhost/nasaaccess"
echo "geoserver_user: admin"
echo "geoserver_password: geoserver"
data_path: /home/internship/apps/tethys_nasaaccess/nasaaccess_data/
nasaaccess_R: /home/internship/miniforge3/envs/tethys/bin/Rscript
nasaaccess_script: /home/internship/apps/tethys_nasaaccess/tethysapp/nasaaccess/scripts/nasaaccess.R
geoserver_workspace: nasaaccess
geoserver_URI: http://localhost/nasaaccess
geoserver_user: admin
geoserver_password: geoserver
```

To install NASAaccess into Tethys, paste the following lines:

```
cd ~/apps/tethys_nasaaccess
PIP_NO_BUILD_ISOLATION=1 tethys install -d -v -w 2>&1 | tee ~/nasaaccess_tethys_install.log
```

When prompted for the service name as ADPC, enter 'primary_geoserver'. For the rest of the required inputs, follow the output of the previous step.

```
******************************
Configuring geoserver_workspace
******************************
Type: CustomSetting
Description: Geoserver Workspace
Required: False

Enter the desired value for the current custom setting: geoserver_workspace
nasaaccess
geoserver_workspace successfully set with value: nasaaccess.

************************
Configuring geoserver_URI
************************
Type: CustomSetting
Description: Geoserver URI
Required: False

Enter the desired value for the current custom setting: geoserver_URI
http://localhost/nasaaccess
geoserver_URI successfully set with value: http://localhost/nasaaccess.

************************
Configuring geoserver_user
************************
Type: CustomSetting
Description: Geoserver User
Required: False

Enter the desired value for the current custom setting: geoserver_user
admin
geoserver_user successfully set with value: admin.

******************************
Configuring geoserver_password
******************************
Type: CustomSetting
Description: Geoserver Password
Required: False

Enter the desired value for the current custom setting: geoserver_password
geoserver
geoserver_password successfully set with value: geoserver.
Services Configuration Completed.
Successfully installed nasaaccess into the active Tethys Portal.

Enter the desired value for the current custom setting: nasaaccess_script
/home/internship/apps/tethys_nasaaccess/tethysapp/nasaaccess/scripts/nasaaccess.R
nasaaccess_script successfully set with value: /home/internship/apps/tethys_nasaaccess/tethysapp/nasa
access/scripts/nasaaccess.R.
```

What-If: the wrong value is pasted during installation?

Stop the installer by clicking CTRL+c, and then reset and reinstall, by running:

```
conda activate tethys
cd ~/apps/tethys_nasaaccess
tethys uninstall nasaaccess -f || true
python -m pip uninstall -y tethysapp-nasaaccess || true
PIP_NO_BUILD_ISOLATION=1 tethys install -d -v -w 2>&1 | tee ~/nasaaccess_tethys_install.log
```

Then enter the correct values. These values can also be fixed after installing NASAaccess and accessing it through Tethys. The values will be found in the ‘settings’ of the app.

What-If: installation fails with: No module named 'tethys_apps'?

This can happen because the NASAaccess app uses older ‘setup.py’ packaging. Install with pip build isolation disabled:

```
conda activate tethys
cd ~/apps/tethys_nasaaccess
PIP_NO_BUILD_ISOLATION=1 tethys install -d -v -w 2>&1 | tee ~/nasaaccess_tethys_install.log
```

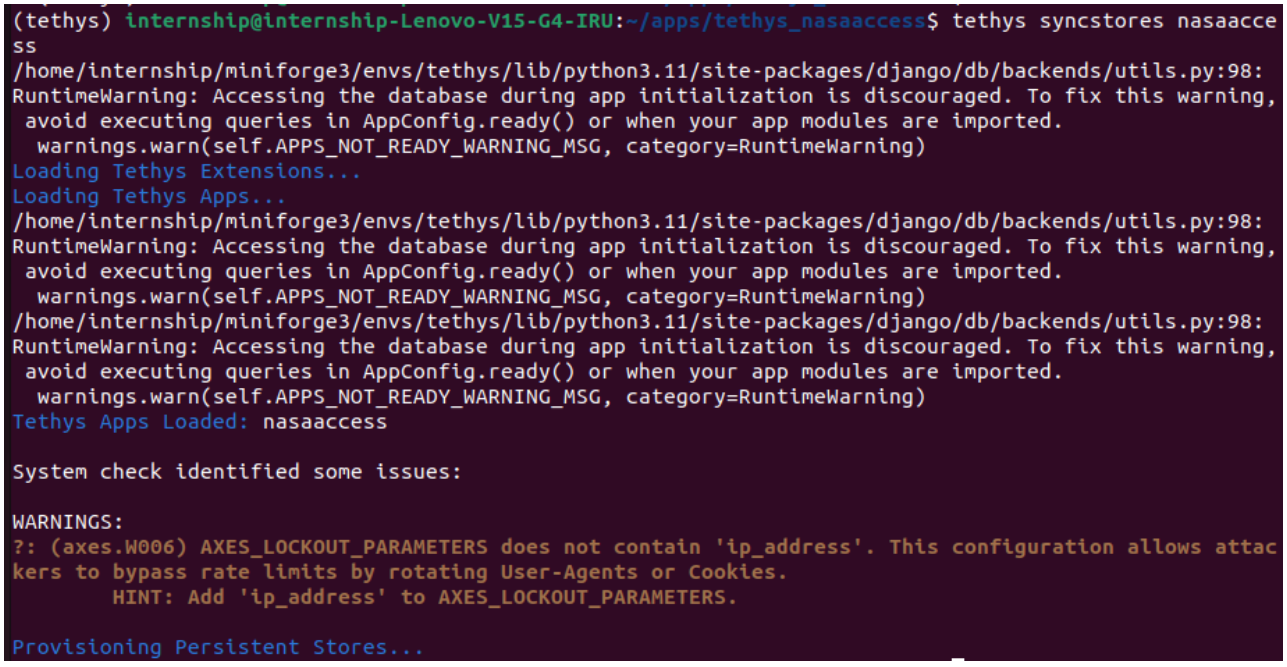
If it still fails, install compatible pip/setuptools versions inside the tethys environment and retry:

```
conda install -n tethys -c conda-forge -y "pip=23.0.1" "setuptools<68" wheel
cd ~/apps/tethys_nasaaccess
PIP_NO_BUILD_ISOLATION=1 tethys install -d -v -w 2>&1 | tee ~/nasaaccess_tethys_install.log
```

11 Sync Stores & Start NASAaccess

Run the following lines together or separately. In the output of the 'syncstores', you must see the 'Tethys Apps loaded: nasaaccess' line.

```
tethys syncstores nasaaccess
tethys start
```



```
(tethys) internship@internship-Lenovo-V15-G4-IRU:~/apps/tethys_nasaaccess$ tethys start
/home/internship/miniforge3/envs/tethys/lib/python3.11/site-packages/django/db/backends/utils.py:98:
RuntimeWarning: Accessing the database during app initialization is discouraged. To fix this warning,
 avoid executing queries in AppConfig.ready() or when your app modules are imported.
  warnings.warn(self.APPS_NOT_READY_WARNING_MSG, category=RuntimeWarning)
Loading Tethys Extensions...
Loading Tethys Apps...
/home/internship/miniforge3/envs/tethys/lib/python3.11/site-packages/django/db/backends/utils.py:98:
RuntimeWarning: Accessing the database during app initialization is discouraged. To fix this warning,
 avoid executing queries in AppConfig.ready() or when your app modules are imported.
  warnings.warn(self.APPS_NOT_READY_WARNING_MSG, category=RuntimeWarning)
/home/internship/miniforge3/envs/tethys/lib/python3.11/site-packages/django/db/backends/utils.py:98:
RuntimeWarning: Accessing the database during app initialization is discouraged. To fix this warning,
 avoid executing queries in AppConfig.ready() or when your app modules are imported.
  warnings.warn(self.APPS_NOT_READY_WARNING_MSG, category=RuntimeWarning)
Tethys Apps Loaded: nasaaccess

/home/internship/miniforge3/envs/tethys/lib/python3.11/site-packages/django/db/backends/utils.py:98:
RuntimeWarning: Accessing the database during app initialization is discouraged. To fix this warning,
 avoid executing queries in AppConfig.ready() or when your app modules are imported.
  warnings.warn(self.APPS_NOT_READY_WARNING_MSG, category=RuntimeWarning)
Loading Tethys Extensions...
Loading Tethys Apps...
/home/internship/miniforge3/envs/tethys/lib/python3.11/site-packages/django/db/backends/utils.py:98:
RuntimeWarning: Accessing the database during app initialization is discouraged. To fix this warning,
 avoid executing queries in AppConfig.ready() or when your app modules are imported.
  warnings.warn(self.APPS_NOT_READY_WARNING_MSG, category=RuntimeWarning)
/home/internship/miniforge3/envs/tethys/lib/python3.11/site-packages/django/db/backends/utils.py:98:
RuntimeWarning: Accessing the database during app initialization is discouraged. To fix this warning,
 avoid executing queries in AppConfig.ready() or when your app modules are imported.
  warnings.warn(self.APPS_NOT_READY_WARNING_MSG, category=RuntimeWarning)
Tethys Apps Loaded: nasaaccess

Performing system checks...

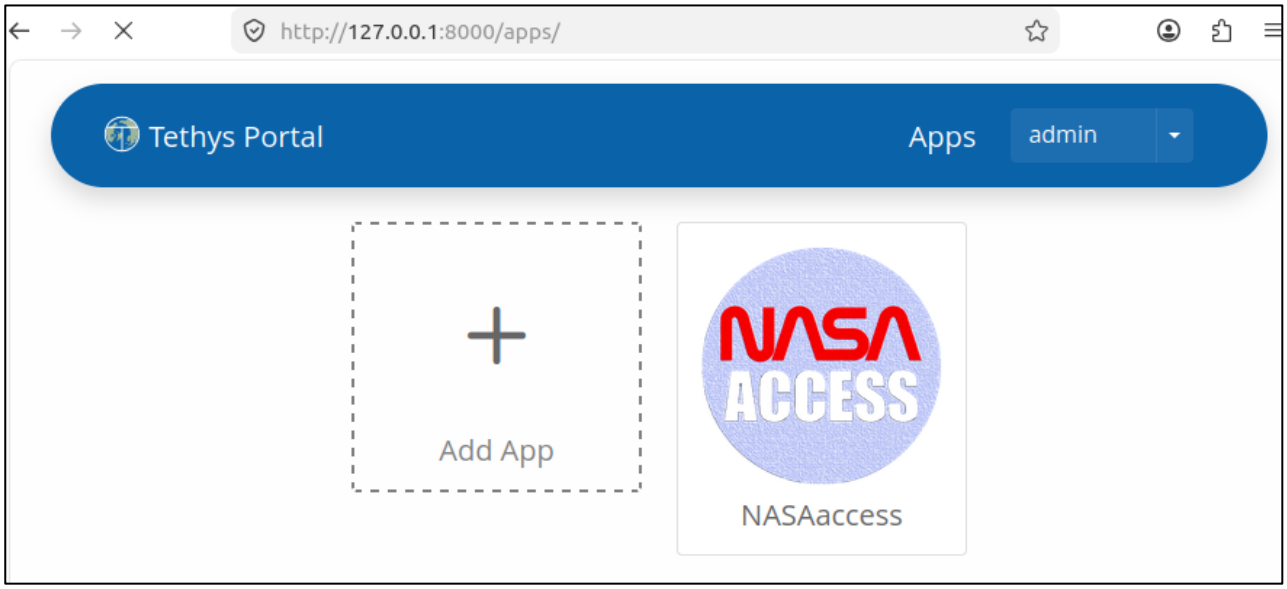
System check identified some issues:

WARNINGS:
?: (axes.W006) AXES_LOCKOUT_PARAMETERS does not contain 'ip_address'. This configuration allows attac
kers to bypass rate limits by rotating User-Agents or Cookies.
        HINT: Add 'ip_address' to AXES_LOCKOUT_PARAMETERS.

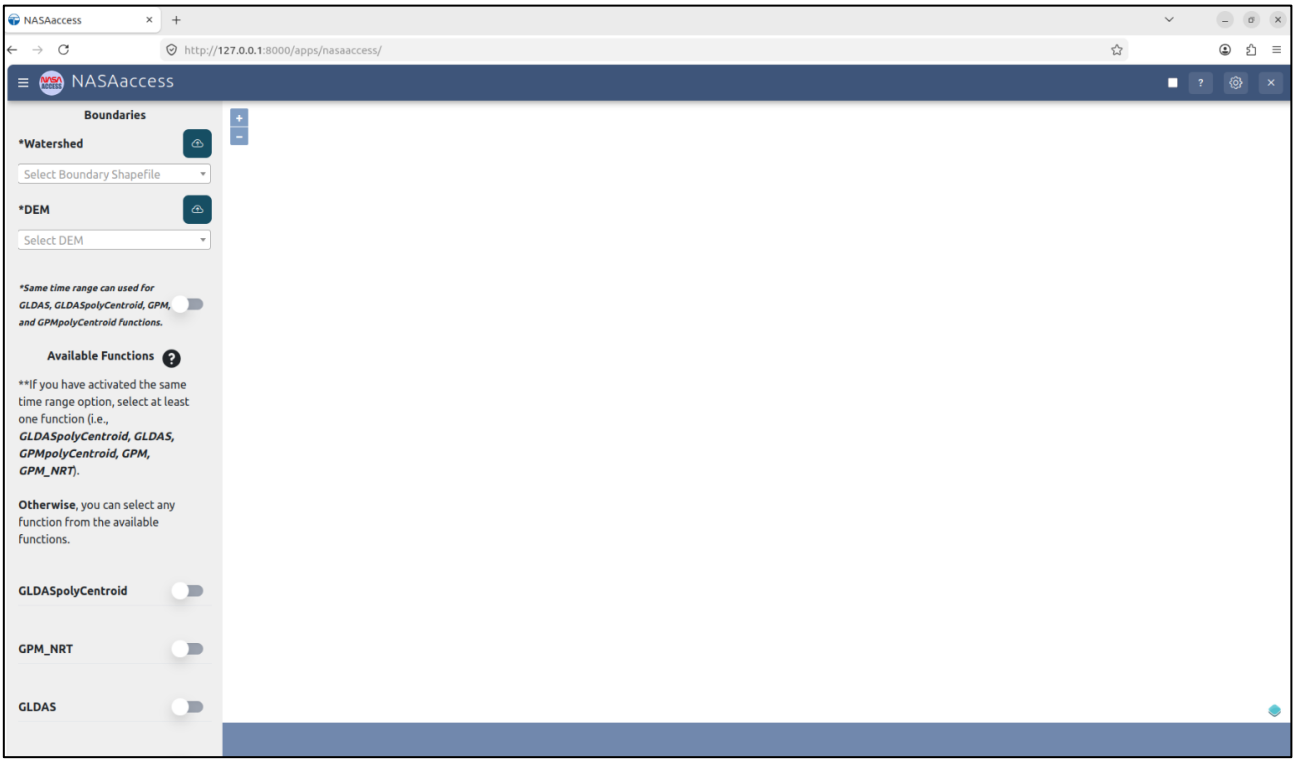
System check identified 1 issue (0 silenced).
June 10, 2026 - 13:55:40
Django version 5.2.15, using settings 'tethys_portal.settings'
Starting ASGI/Daphne version 4.2.2 development server at http://127.0.0.1:8000/
Quit the server with CONTROL-C.
```

Open, the following link in your output by CTRL+click:

http://127.0.0.1:8000/



What-If: the basemap is blank?



Close Tethys using CTRL+c and run the following lines:

```
conda activate tethys
cd ~/apps/tethys_nasaaccess
grep -nE "BingMaps|terrestris|OSM-WMS|baseLayer" tethysapp/nasaaccess/public/js/main.js | head -40
```

If you see this:

```
ol.source.BingMaps
```

```
(tethys) internship@internship-Lenovo-V15-G4-IRU:~/apps/tethys_nasaaccess$ conda activate tethys
cd ~/apps/tethys_nasaaccess
grep -nE "BingMaps|terrestris|OSM-WMS|baseLayer" tethysapp/nasaaccess/public/js/main.js | head -40
39:    baseLayer,
184:    baseLayer = new ol.layer.Tile({
185:      source: new ol.source.BingMaps({
203:    layers = [baseLayer];
```

Then your app is still using the default Bing basemap. To fix this, first stop Tethys using CTRL+c. Back up the original JavaScript file by running:

```
cd ~/apps/tethys_nasaaccess
cp tethysapp/nasaaccess/public/js/main.js tethysapp/nasaaccess/public/js/main.js.bak_bing_$(date
+%Y%m%d_%H%M%S)
```

Patch the map by running these lines together:

```
python - <<'PY'
from pathlib import Path
import re
p = Path("tethysapp/nasaaccess/public/js/main.js")
s = p.read_text()

new_block = '''    baseLayer = new ol.layer.Image({
      source: new ol.source.ImageWMS({
        url: "https://ows.terrestris.de/osm/service?",
        params: {
          "LAYERS": "OSM-WMS",
          "VERSION": "1.1.1",
          "FORMAT": "image/png"
        },
        ratio: 1
      }),
      title: "baselayer",
    });
    layers = [baseLayer];'''
pattern =
r'\s*baseLayer\s*=\s*new\s+ol\.layer\.(?:Tile|Image)\(\{.*?\n\s*\}\);\s*\n\s*layers\s*=\s*\[baseLayer\];'
s2, n = re.subn(pattern, "\n" + new_block, s, count=1, flags=re.S)
if n != 1:
    raise SystemExit(
        "Could not automatically replace the basemap block. "
        "Inspect the file with: sed -n '175,215p' tethysapp/nasaaccess/public/js/main.js"
    )
p.write_text(s2)
print("Basemap replaced with OpenStreetMap WMS.")
PY
```

```
(tethys) internship@internship-Lenovo-V15-G4-IRU:~/apps/tethys_nasaaccess$ python - <<'PY'
from pathlib import Path
import re
p = Path("tethysapp/nasaaccess/public/js/main.js")
s = p.read_text()
new_block = '''baseLayer = new ol.layer.Image({
    source: new ol.source.ImageWMS({
        url: "https://ows.terrestris.de/osm/service?",
        params: {
            "LAYERS": "OSM-WMS",
            "VERSION": "1.1.1",
            "FORMAT": "image/png"
        },
        ratio: 1,
        crossOrigin: "anonymous"
    }),
    title: "baselayer",
});'''
pattern = r'baseLayer\s*=\s*new\s+ol\.layer\.Tile\(\{\s*source:\s*new\s+ol\.source\.BingMaps\(\{.*?\}
\),\s*title:\s*"baselayer",\s*\}\);'
s2, n = re.subn(pattern, new_block, s, count=1, flags=re.S)
if n != 1:
    raise SystemExit(
        "Could not find the Bing basemap block. Run this command to inspect the file:\n"
        "grep -nE 'BingMaps|baseLayer|terrestris|OSM-WMS' tethysapp/nasaaccess/public/js/main.js | he
ad -40"
    )
p.write_text(s2)
print("Basemap changed from BingMaps to OpenStreetMap WMS.")
PY
Basemap changed from BingMaps to OpenStreetMap WMS.
```

If the patch command says ‘Could not automatically replace the basemap block’, but the grep command already shows terrestris or OSM-WMS, the patch has already been applied. Continue

with ‘tethys manage collectstatic --clear –noinput’, restart Tethys, and hard-refresh Firefox, as instructed below. If you want to confirm the patch, run the following line:

```
grep -nE "BingMaps|terrestris|OSM-WMS|baseLayer" tethysapp/nasaaccess/public/js/main.js | head -40
```

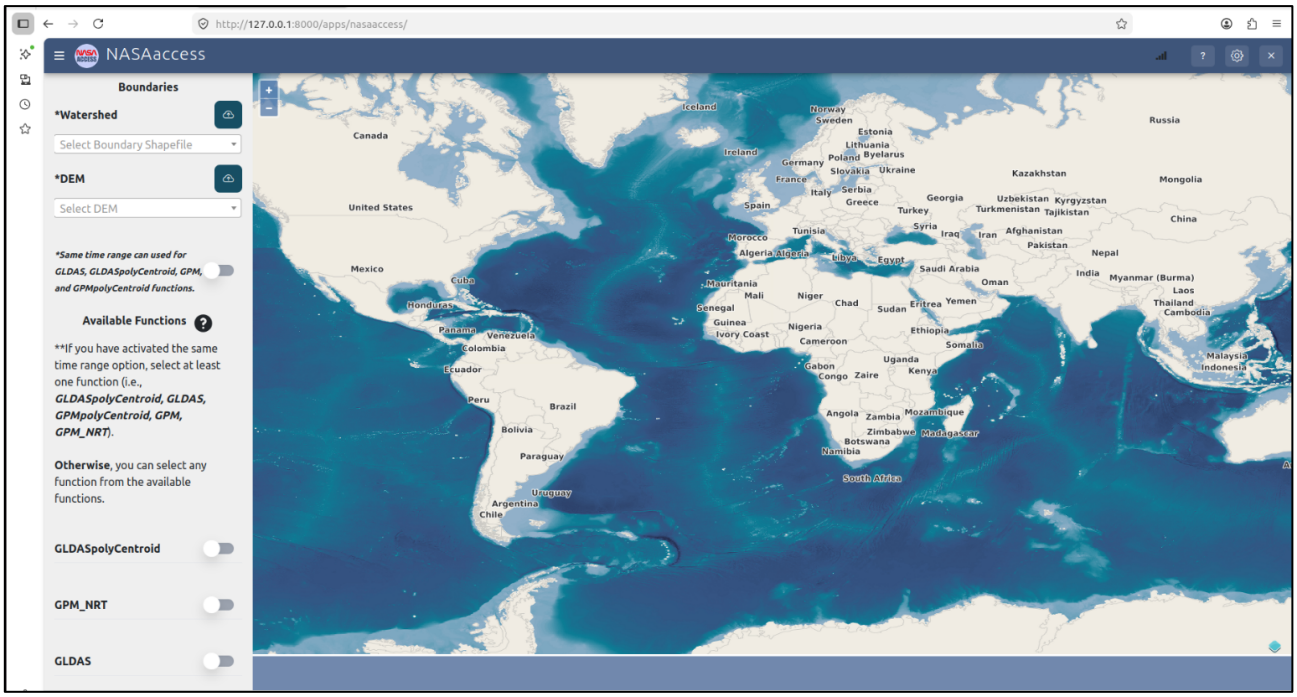
You should see ‘terrestris’ or ‘OSM-WMS’, and not ‘Bing Maps’.

```
(tethys) internship@internship-Lenovo-V15-G4-IRU:~/apps/tethys_nasaaccess$ grep -nE "BingMaps|terrest
ris|OSM-WMS|baseLayer" tethysapp/nasaaccess/public/js/main.js | head -40
39:    baseLayer,
184:    baseLayer = new ol.layer.Image({
186:        url: "https://ows.terrestris.de/osm/service?",
188:            "LAYERS": "OSM-WMS",
209:    layers = [baseLayer];
```

Next step is to clear Tethys static files and restart:

```
tethys manage collectstatic --clear --noinput
tethys start
```

For a hard refresh, in NASAaccess page, click on F12 on your keyboard → Go to ‘Network’ → check Disable Cache → reload the page by clicking on CTRL+SHIFT+r. The basemap should now display.



12 Verify the Full Installation

Open a new Terminal, leaving Tethys running. First, activate your environment by running:

```
conda activate tethys
```

Check Docker:

```
docker --version
docker compose version
docker ps
```

Check Python, Django, and Tethys:

```
python --version
python -c "import django; print(django.get_version())"
tethys version
```

Check R and NASAaccess:

```
R --version | head -n 1
which Rscript
Rscript -e "library(NASAaccess); print(packageVersion('NASAaccess'))"
```

Check NASAaccess app package:

```
python -m pip show tethysapp-nasaaccess
```

```
(base) internship@internship-Lenovo-V15-G4-IRU:~$ conda activate tethys
(tethys) internship@internship-Lenovo-V15-G4-IRU:~$ docker --version
docker compose version
docker ps
Docker version 29.5.3, build d1c06ef
Docker Compose version v5.1.4
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
(tethys) internship@internship-Lenovo-V15-G4-IRU:~$ python --version
python -c "import django; print(django.get_version())"
tethys version
Python 3.11.15
5.2.15
4.5.2
(tethys) internship@internship-Lenovo-V15-G4-IRU:~$ R --version | head -n 1
which Rscript
Rscript -e "library(NASAaccess); print(packageVersion('NASAaccess'))"
R version 4.5.3 (2026-03-11) -- "Reassured Reassurer"
/home/internship/miniforge3/envs/tethys/bin/Rscript
[1] ‘4.2.0’
(tethys) internship@internship-Lenovo-V15-G4-IRU:~$ python -m pip show tethysapp-nasaaccess
Name: tethysapp-nasaaccess
Version: 2.0.1
Summary: Web interface for accessing and visualizing climate and weather data from NASA's Earth Obser
ving System Data and Information System (EOSDIS)
Home-page: https://github.com/imohamme/tethys_nasaaccess
Author: Spencer McDonald, Ibrahim Mohammed, Giovanni Romero
Author-email: ibrahim.mohammed@nasa.gov
License: NASA OPEN SOURCE AGREEMENT VERSION 1.3
Location: /home/internship/apps/tethys_nasaaccess
Editable project location: /home/internship/apps/tethys_nasaaccess
Requires:
Required-by:
```

Check GeoServer:

```
curl -s -o /dev/null -w "%{http_code}\n" http://127.0.0.1:8181/geoserver/web/
```

Expected:

```
200
```

Check Tethys:

```
curl -s -o /dev/null -w "%{http_code}\n" http://127.0.0.1:8000/
```

Expected:

```
200
```

Or:

```
302
```

Check NASAaccess:

```
curl -s -o /dev/null -w "%{http_code}\n" http://127.0.0.1:8000/apps/nasaaccess/
```

Expected:

```
200
```

Or:

```
302
```

What-If: 000 is returned?

If curl returns 000, it means the service is not reachable. For port 8181, GeoServer is might be stopped. For port 8000, Tethys is probably stopped. Try running:

```
conda activate tethys
sudo systemctl start docker
tethys docker start -c geoserver
docker ps
```

```
(tethys) internship@internship-Lenovo-V15-G4-IRU:~$ curl -s -o /dev/null -w "%{http_code}\n" http://1
27.0.0.1:8181/geoserver/web/
000
(tethys) internship@internship-Lenovo-V15-G4-IRU:~$ conda activate tethys
sudo systemctl start docker
tethys docker start -c geoserver
docker ps
[sudo] password for internship:
Starting GeoServer container...
CONTAINER ID   IMAGE                              COMMAND                  CREATED      STATUS
          PORTS
                    NAMES
bb45d6ad0890   tethysplatform/geoserver:latest    "/bin/sh -c $GEOSERV…"   7 days ago   Up Less than a
 second   0.0.0.0:8081-8084->8081-8084/tcp, [::]:8081-8084->8081-8084/tcp, 0.0.0.0:8181->8181/tcp, [:
:]:8181->8181/tcp   tethys_geoserver
```

If it works, close the second terminal and restart Tethys by CTRL+c and then running ‘tethys start’ again.

13 Upload Required Input Files

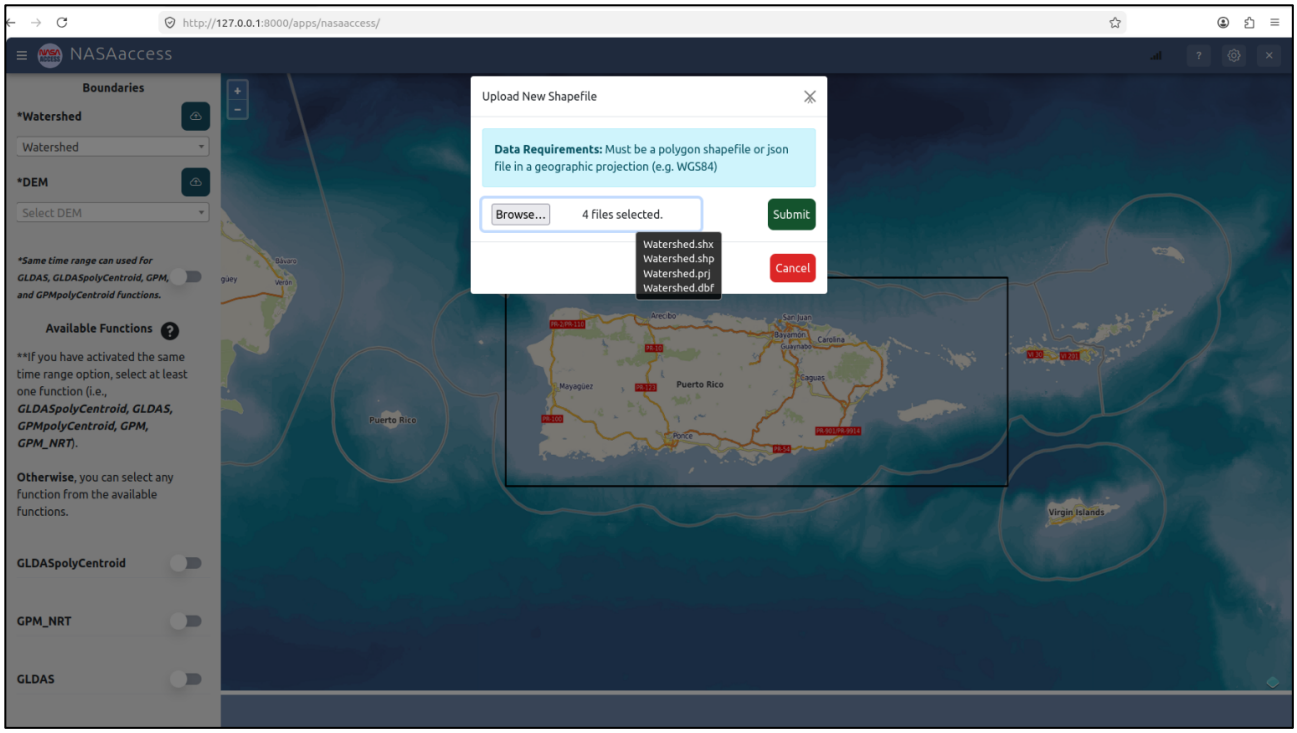
13.1 Watershed Boundary

Upload a polygon shapefile with exactly these file extensions selected together:

```
watershed.shp
watershed.shx
watershed.dbf
watershed.prj
```

The four files are part of the watershed shapefile, each saving unique and important data of the shapefile itself. Hence, make sure they have the same name, with only the extension is different.

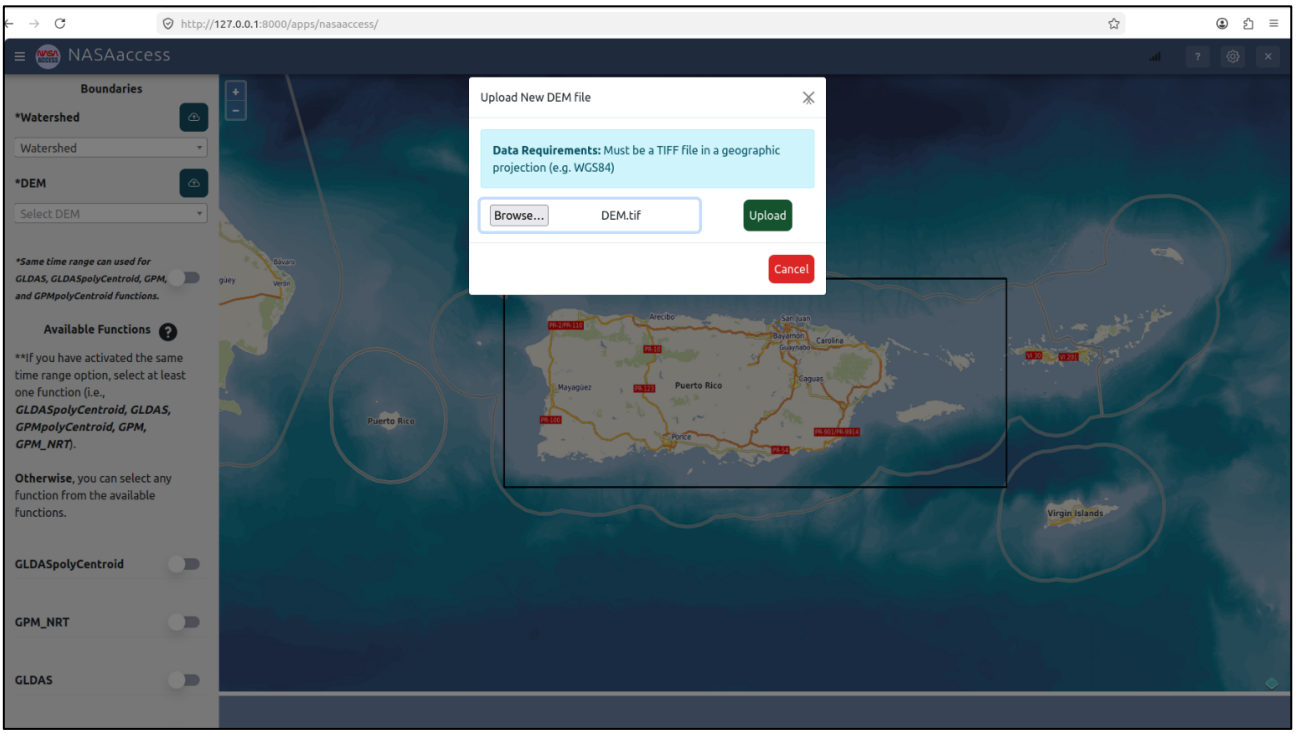
The watershed must be a polygon shapefile in the geographic coordinate system, WGS84 / EPSG:4326. Do not upload a zipped shapefile.



13.2 DEM

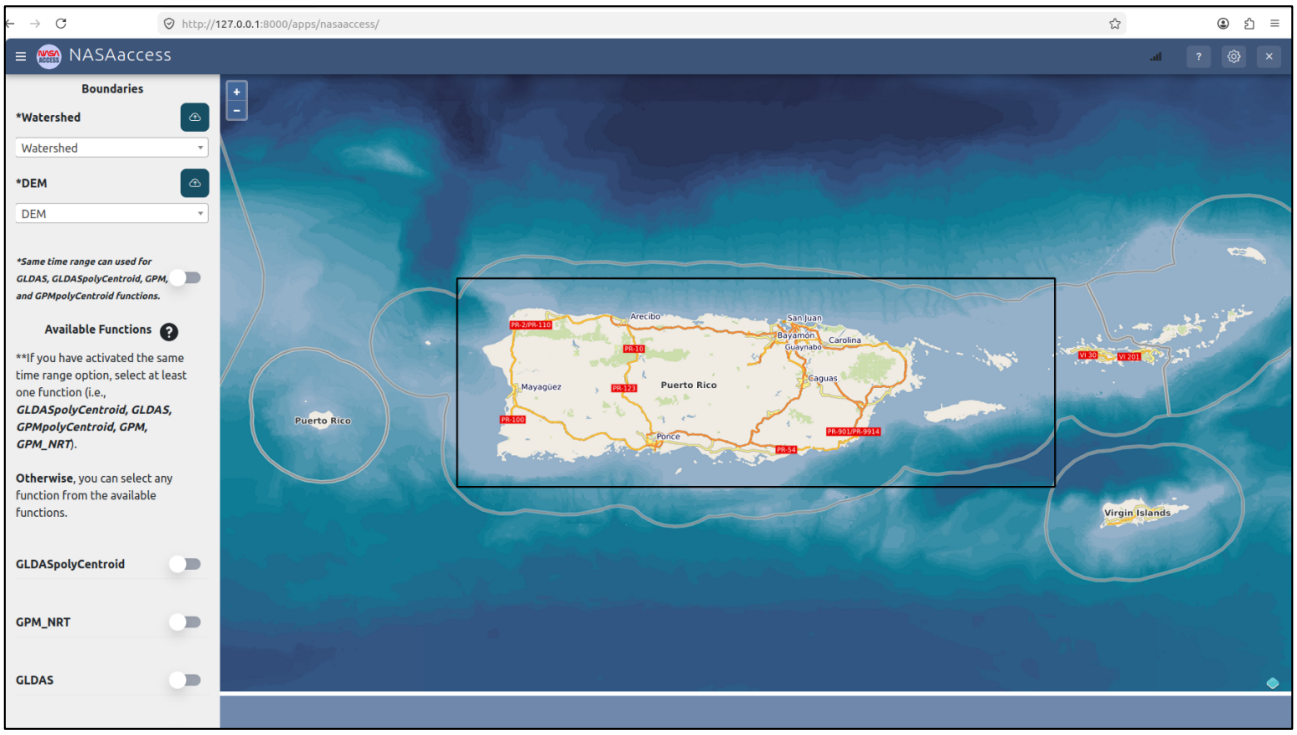
Upload a DEM as:

dem.tif



What-If: DEM upload says it is already in GeoServer?

First refresh the NASAaccess page by clicking CTRL+r. Then, check the DEM dropdown. If the DEM appears, select it.

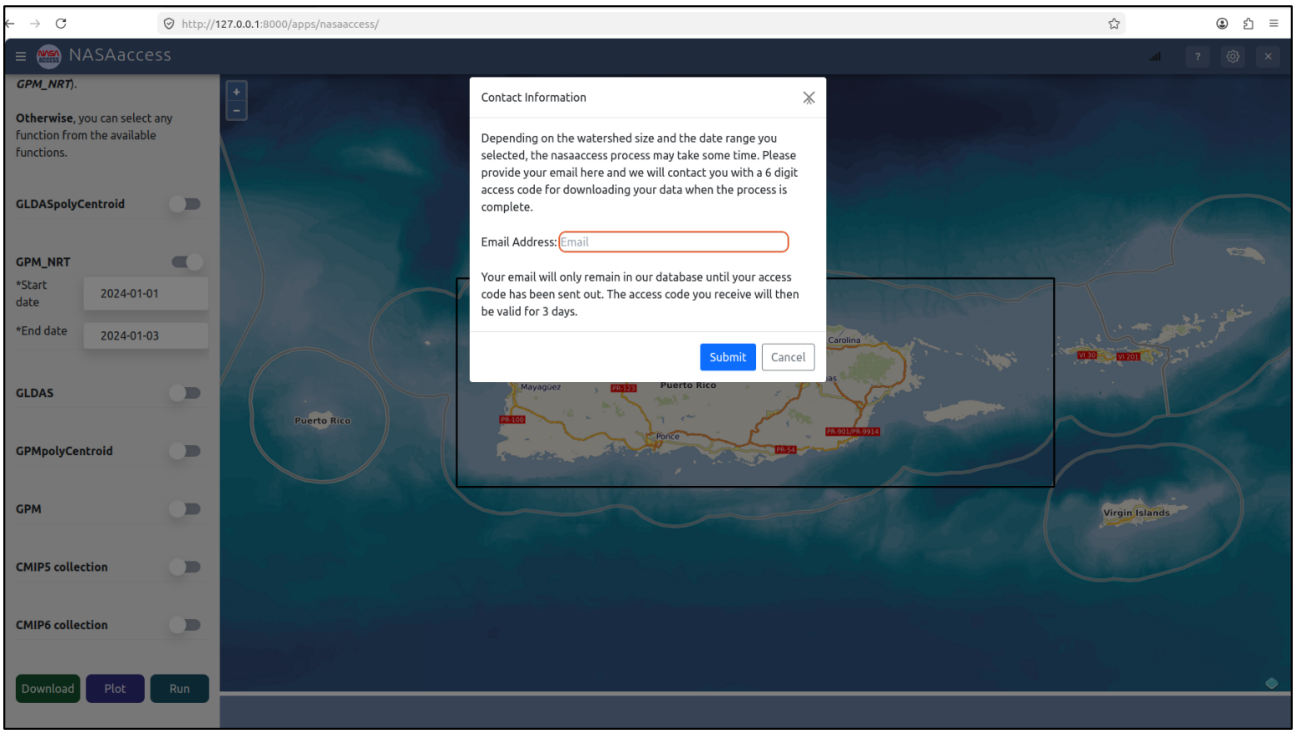


14 Run A Small Test Job

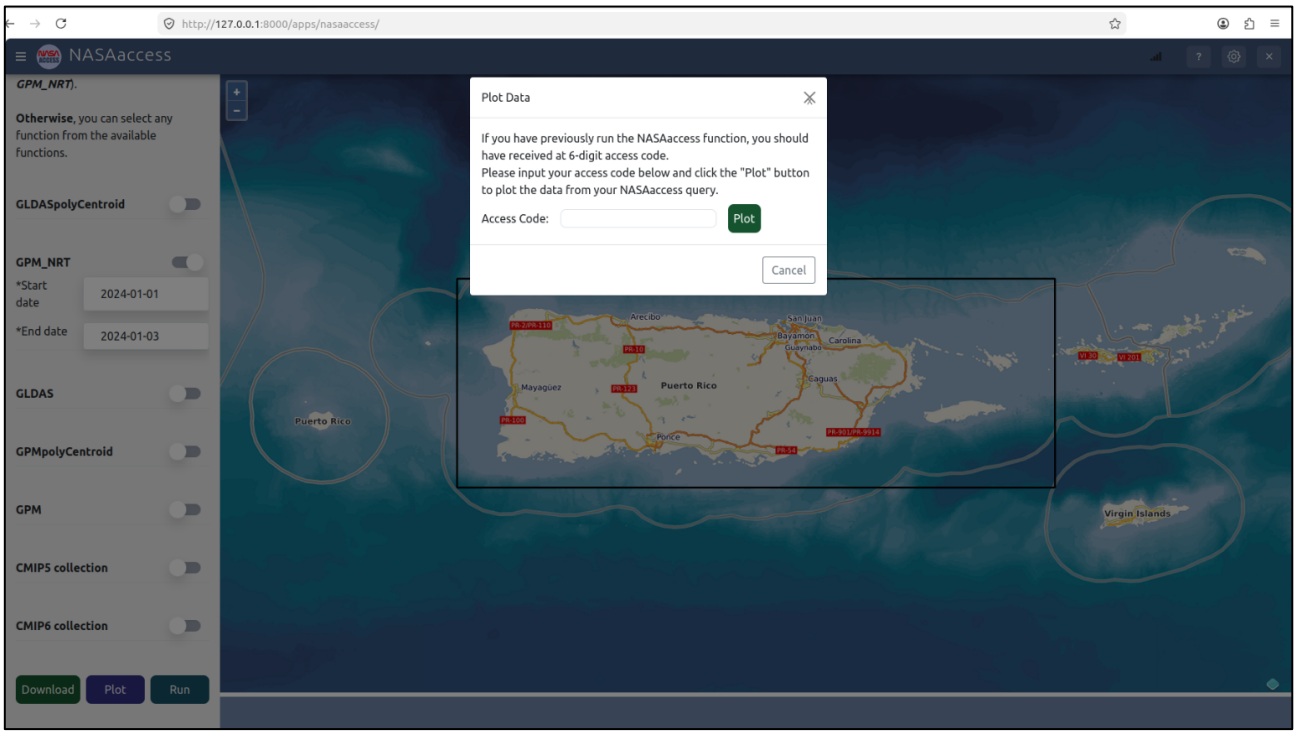
Let's do a test run to make sure everything works. First, do not select every function for the first test. Start by selecting:

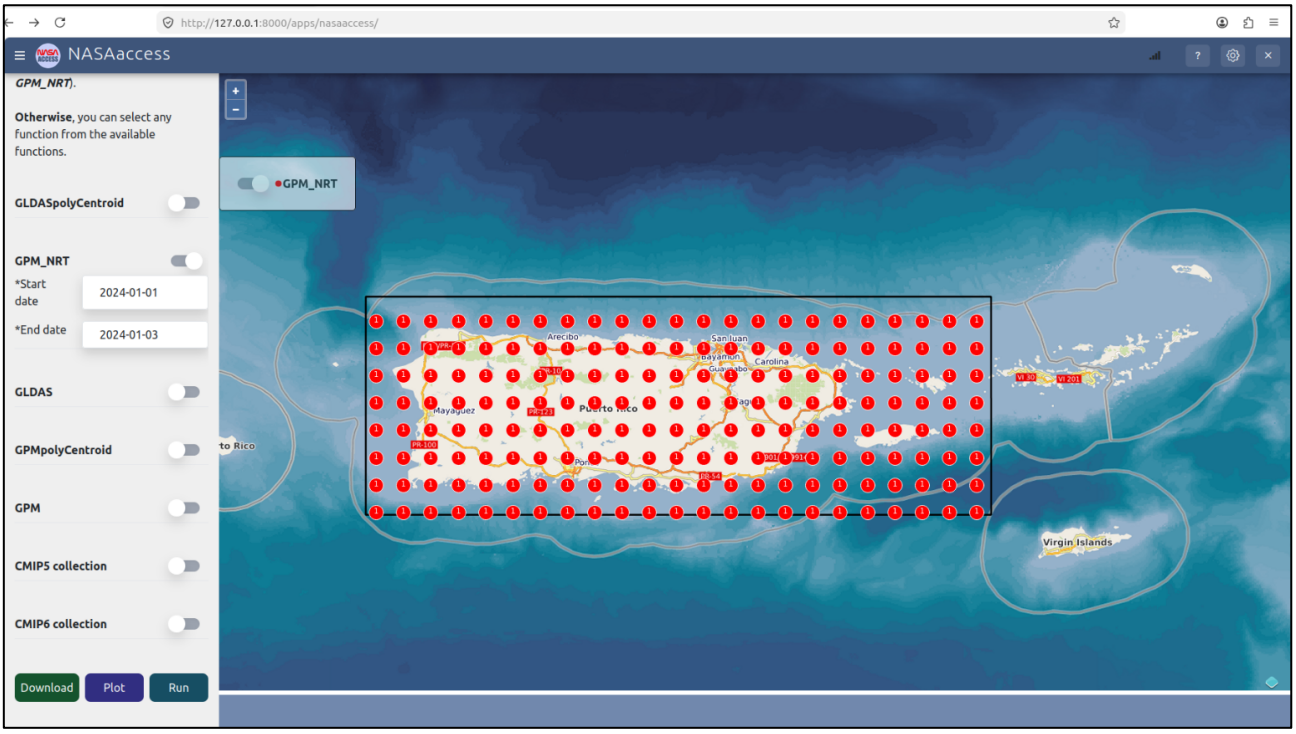
Function: GPM_NRT Start date: 2024-01-01 End date: 2024-01-03

Click on 'Run' below. You will be prompted to enter an email that code will be sent to. Enter any email you wish to receive the code in.

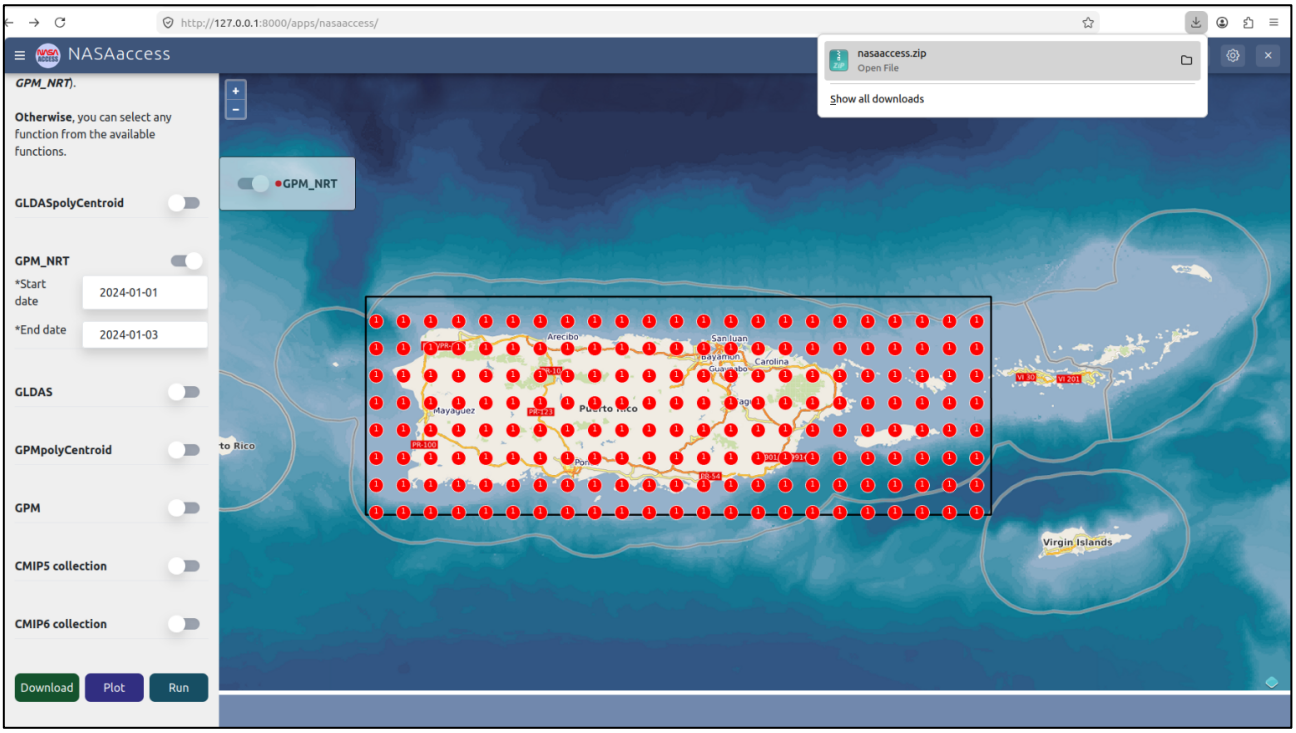


In a few minutes, you will receive an email notification. Copy the access code you got, and in the NASAaccess page click on ‘Plot’. A window will open to paste the code you received in. Paste the code and wait.





Click on 'Download' to download the .zip file with all the results.



15 Daily Startup After Installation

After installation, you can start GeoServer and Tethys by running:

```
conda activate tethys
cd ~/apps/tethys_nasaaccess
tethys docker start -c geoserver
tethys start
```